

Jan. 8, 1946.

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2,392,745

ESCAPE AND BALANCE STAFF

Filed Jan. 21, 1943

FIG. 1.

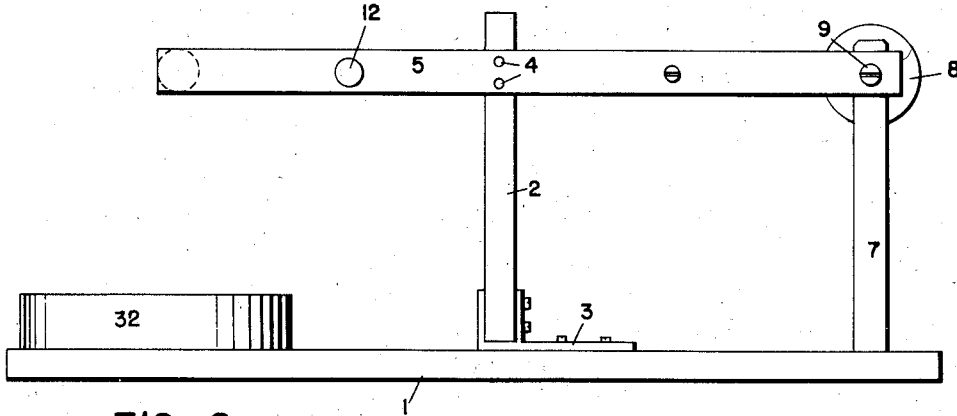


FIG. 2.

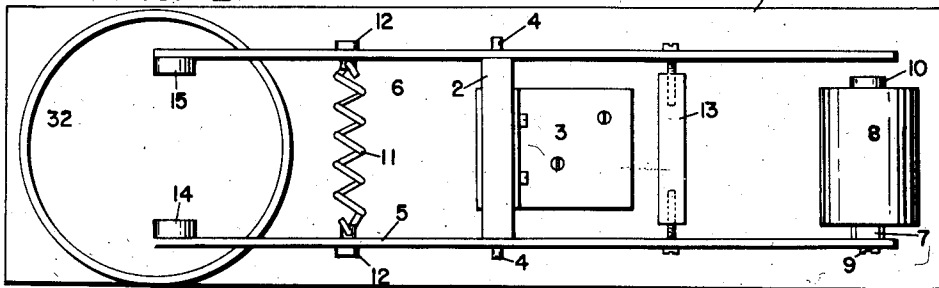


FIG. 3.

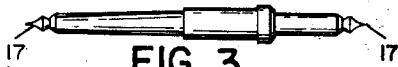
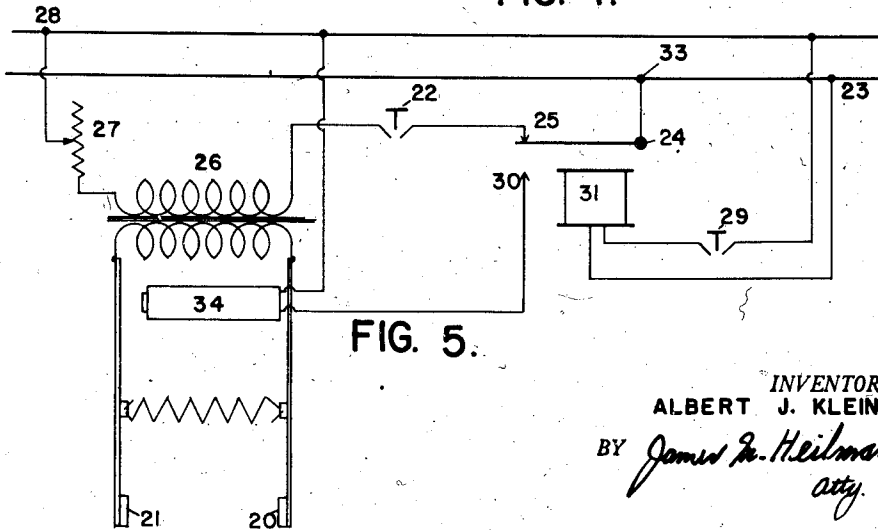


FIG. 4.



FIG. 5.



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2,392,745

ESCAPE AND BALANCE STAFF

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Application January 21, 1943, Serial No. 473,076

2 Claims. (Cl. 58—140)

This invention relates to the production of the staff such as an escape or balance staff for high-grade timepieces.

Escape staffs for high-grade timepieces have pivots about 7 thousandths of an inch in diameter and must support a wheel which must withstand the shocks and pressures involved in the operation of the escapement. This produces tremendous strain on the small pivot ends and the degree of hardness of said ends is extremely important in maintaining the timepiece in accurate running condition. It is necessary to have an absolutely true staff so that the wheel and pinion will run accurately. To assure this accuracy the shafts must be absolutely true and must, therefore, be turned or ground to proper dimensions. The machinery necessary for grinding such a part is impractical which leaves the need for turning said parts to the desired accuracy.

In order to turn the parts it is necessary to have the metal at the proper degree of hardness which is too soft to give the desired hardness to the pivot ends. Accordingly, this invention is directed to the production of a staff, the pivot ends of which are hardened without warping or distorting the center portion and without hardening the center portion so that it cannot be turned.

The object of this invention is to provide a staff which has a relative soft body portion and relative hard end portions.

A further object is to produce a finished product which because of its variation of temper throughout its length fulfills the need of a staff having toughness and workability in its body portion and hardness and the quality of taking a high polish in its ends.

The invention is illustrated in the accompanying drawing in which

Fig. 1 is a front plan of an apparatus designed to carry out the method hereinafter described.

Fig. 2 is a top plan of the same apparatus.

Fig. 3 is an enlarged view of one of the finished parts upon which the method is worked.

Fig. 4 is an enlarged detail view of the end of one of the parts.

Fig. 5 is a diagrammatic view including the electrical circuit.

The finished product which is the invention here submitted may be made by various methods and with different apparatus, the method and apparatus here described being convenient and thorough.

This apparatus consists of a base 1 having centrally mounted thereon an upright 2 formed of

insulating material and firmly secured to the base by support 3. A pair of pins 4 extending outward from near the top of the upright carries metallic arms 5 and 6 which are fitted loosely enough to said pins to permit movement toward and away from each other in a plane parallel to the base. A second upright arm 7 carries a solenoid 8 arranged in the plane of the metallic arms 5 and 6. Arm 5 is secured to said upright arm 7 at 9 while arm 6 is free to move under the attraction of the solenoid core 10 in one direction and the pull of a spring 11 supported in insulated plugs 12 in the other direction. This movement is, of course, slight and is limited by said solenoid core and an adjustable stop 13. Attached to the free end of the metallic arm are silver contacts 14 and 15 while located on the base directly below the contacts is a container 32 designed to hold quenching oil or water.

In the making of small watch parts, the staffs such as the escape pinion and the balance staff, the size of the supporting ends or pivots 17 (Fig. 3) is extremely small when compared with the gear or wheel designed to be carried by said staffs. It is, therefore, necessary that the machining of said staffs be extremely accurate. It is also necessary that the ends 17 be of a certain hardness to prevent wear, and distortion due to shock. If the entire staff is hardened to this desired degree, the length is warped or distorted and the extreme hardness makes it impossible to machine the staffs to the exact trueness required. To obtain this trueness it would be necessary to grind them which operation is impractical both as to time and material. It is, therefore, desirable to harden just the ends of the staff without affecting the central portion.

The ends of the staffs are roughed out to the shape shown in Fig. 4 and then a groove 18 is cut in each end, the extreme ends 19 (Fig. 4) being unfinished. The ends 19 are then heated by resistance by passing an electric current through the narrowed portion formed by groove 18, causing the heat to be concentrated, and the entire ends becoming heated to the desired temperature necessary to develop the proper structure. The staffs are then dropped in a quenching bath and later reheated to give the desired temper, retaining just the right amount of hardness in the ends, after which the ends are ground and polished to the final shape shown in dotted lines in Fig. 4.

The method applied to the apparatus here shown is as follows:

Referring to Fig. 5 a staff having its end formed

as in Fig. 4 is placed between the electrodes 20 and 21. The switch 22 is depressed, closing the circuit from contact 33 in the 110 volt line through contact 25, through the primary coil of the transformer 26, the controlled resistance 27 to the line at 28. The secondary of the transformer supplies the current to the contacts 20 and 21 and heats the staff carried between said contacts. Upon the staff reaching the desired heat which is determined by observation, switch 29 is depressed breaking the circuit at 25 and establishing a new circuit at 30 to operate solenoid 34, causing the contacts 20 and 21 to be more widely separated, dropping the staff into a quenching bath. The breaking of the heating circuit momentarily before dropping the staff prevents arcing and the formation of undesired pits on the staff ends.

After cooling, the staffs are finished by tempering, then grinding to the desired form.

What is claimed is:

- 5 1. A monometallic integral escape staff for timepieces having a soft workable body portion and pivot ends sufficiently hard to break before bending.
- 10 2. A monometallic integral escape staff for timepieces formed with a body portion of comparative cylindrical shape reduced portions adjacent said body portion and pivot ends beyond said reduced portions, that part of the staff included between the ends and the reduced portion being integral with said body portion and of a
15 hardness sufficient to break before bending.

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